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A certain state's milk production was 980 million pounds in 2007 and 2.7 billion pounds in 2014. Approximately how many more million gallons of milk did the state produce in 2014 than in 2007? (1 billion = 10^9 and 1 gallon = 8.6 pounds.)

A. 100

B. 200

C. 1,700

D. 8,200

E. 14,800

•

Working simultaneously and independently at an identical constant rate, four machines of a certain type can produce a total of x units of Product P in 6 days. How many of these machines, working simultaneously and independently at this constant rate, can produce a total of $3x$ units of Product P in 4 days?

A. 24

B. 18

C. 16

D. 12

E. 8

•

The symbol Δ denotes one of the four arithmetic operations: addition, subtraction, multiplication, or division. If 6 Δ 3 ≤ 3, which of the following must be true?

I. 2 Δ 2 = 0

II. 2 Δ 2 = 1

III. 4 Δ 2 = 2

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

•

If $mn \neq 0$ and 25 percent of n equals $37\frac{1}{2}$ percent of m, what is the value of $\frac{12n}{m}$?

A. 18

B. $\frac{32}{3}$

C. 8

D. 3

E. $\frac{9}{8}$

•

N and M are each 3-digit integers. Each of the numbers 1, 2, 3, 6, 7, and 8 is a digit of either N or M. What is the smallest possible positive difference between N and M ?

A. 29

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A. 29

B. 49

C. 58

D. 113

E. 131

Technique	Percent of Consumers
Television ads	35%
Coupons	22%
Store displays	18%
Samples	15%

•

The table shows partial results of a survey in which consumers were asked to indicate which one of six promotional techniques most influenced their decision to buy a new food product. Of those consumers who indicated one of the four techniques listed, what fraction indicated either coupons or store displays?

A. $\frac{2}{7}$

B. $\frac{1}{3}$

C. $\frac{2}{5}$

D. $\frac{4}{9}$

E. $\frac{1}{2}$

•

If 65 percent of a certain firm's employees are full-time and if there are 5,100 more full-time employees than part-time employees, how many employees does the firm have?

A. 8,250

B. 10,200

C. 11,050

D. 16,500

E. 17,000

•

The cost C, in dollars, to remove p percent of a certain pollutant from a pond is estimated by using the formula $C = \frac{100,000p}{100-p}$. According to this estimate, how much more would it cost to remove 90 percent of the pollutant from the pond than it would cost to remove 80 percent of the pollutant?

A. $500,000

B. $100,000

C. $50,000

D. $10,000

E. $5,000

•

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If $xy \neq 0$ and $x^2y^2 - xy = 6$, which of the following could be y in terms of x?

I. $\frac{1}{2x}$

II. $-\frac{2}{x}$

III. $\frac{3}{x}$

A. I only

B. II only

C. I and II

D. I and III

E. II and III

•

If the positive number d is the standard deviation of n, k, and p, then the standard deviation of $n+1$, $k+1$, and $p+1$ is

A. $d+3$

B. $d+1$

C. $6d$

D. $3d$

E. d

•

In a certain high school, 80 percent of the seniors are taking calculus, and 60 percent of the seniors who are taking calculus are also taking physics. If 10 percent of the seniors are taking neither calculus nor physics, what percent of the seniors are taking physics?

A. 40%

B. 42%

C. 48%

D. 58%

E. 80%

•

If the units digit of $\frac{5,610.37}{10^k}$ is 6, what is the value of k?

A. 3

B. 2

C. 1

D. −1

E. −2

•

Three printing presses, R, S, and T, working together at their respective constant rates, can do a certain printing job in 4 hours. S and T, working together at their respective constant rates, can do the same job in 5 hours. How many hours would it take R, working alone at its constant rate, to do the same job?

A. 8

B. 10

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B. 10

C. 12

D. 15

E. 20

Results of a Poll

Company	Number Who Own Stock in the Company
AT&T	30
IBM	48
GM	54
FORD	75
US Air	83

•

In a poll, 200 subscribers to *Financial Magazine* X indicated which of five specific companies they own stock in. The results are shown in the table above. If 15 of the 200 own stock in both IBM and AT&T, how many of those polled own stock in neither company?

A. 63

B. 93

C. 107

D. 122

E. 137

•

The sum of all the integers k such that $-26 < k < 24$ is

A. 0

B. –2

C. –25

D. –49

E. –51

•

The number line shown contains three points R, S, and T, whose coordinates have absolute values r, s, and t, respectively. Which of the following equals the average (arithmetic mean) of the coordinates of the points R, S, and T?

A. s

B. $s + t - r$

C. $\frac{r - s - t}{3}$

D. $\frac{r + s + t}{3}$

E. $\frac{s + t - r}{3}$

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•

Mark and Ann together were allocated n boxes of cookies to sell for a club project. Mark sold 10 boxes less than n and Ann sold 2 boxes less than n. If Mark and Ann have each sold at least one box of cookies, but together they have sold less than n boxes, what is the value of n?

A. 11

B. 12

C. 13

D. 14

E. 15

$$\begin{array}{r} 3P5 \\ +4\,\mathrm{QR} \\ \hline 8S4 \end{array}$$

•

In the correctly worked addition problem shown, P, Q, R, and S are digits. If $Q = 2P$, which of the following could be the value of S?

A. 3

B. 4

C. 5

D. 7

E. 9

•

A certain high school has 5,000 students. Of these students, x are taking music, y are taking art, and z are taking both music and art. How many students are taking neither music nor art?

A. $5,000 - z$

B. $5,000 - x - y$

C. $5,000 - x + z$

D. $5,000 - x - y - z$

E. $5,000 - x - y + z$

•

Each person who attended a company meeting was either a stockholder in the company, an employee of the company, or both. If 62 percent of those who attended the meeting were stockholders and 47 percent were employees, what percent were stockholders who were NOT employees?

A. 34%

B. 38%

C. 45%

D. 53%

E. 62%

•

If M is the least common multiple of 90, 196, and 300, which of the following is <u>not</u> a factor of M?

A. 600

B. 700

C. 900

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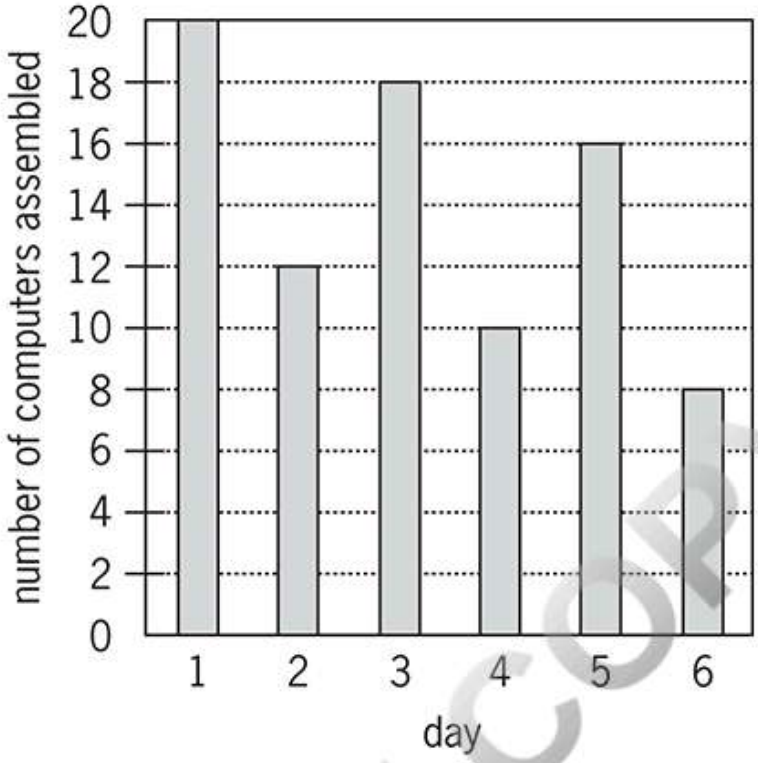
D. 2,100

E. 4,900



•

The graph shows the number of computers assembled during each of 6 consecutive days. From what day to the next day was the percent change in the number of computers assembled the greatest in magnitude?

A. From Day 1 to Day 2

B. From Day 2 to Day 3

C. From Day 3 to Day 4

D. From Day 4 to Day 5

E. From Day 5 to Day 6

•

If $n = 20! + 17$, then n is divisible by which of the following?

I. 15

II. 17

III. 19

A. None

B. I only

C. II only

D. I and II

E. II and III

•

Exchange Rates in a Particular Year

$1 = 5.3 francs

$1 = 1.6 marks

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An American dealer bought a table in Germany for 480 marks and sold the same table in France for 2,385 francs. What was the dealer's gross profit on the two transactions in dollars?

A. $0

B. $50

C. $100

D. $150

E. $200

•

The product of two negative numbers is 160. If the lesser of the two numbers is 4 less than twice the greater, what is the greater number?

A. –20

B. –16

C. –10

D. –8

E. –4

•

According to a certain estimate, the depth $N(t)$, in centimeters, of the water in a certain tank at t hours past 2:00 in the morning is given by $N(t) = -20(t-5)^2 + 500$ for $0 \leq t \leq 10$. According to this estimate, at what time in the morning does the depth of the water in the tank reach its maximum?

A. 5:30

B. 7:00

C. 7:30

D. 8:00

E. 9:00

•

After driving to a riverfront parking lot, Bob plans to run south along the river, turn around, and return to the parking lot, running north along the same path. After running 3.25 miles south, he decides to run for only 50 minutes more. If Bob runs at a constant rate of 8 minutes per mile, how many miles farther south can he run and still be able to return to the parking lot in 50 minutes?

A. 1.5

B. 2.25

C. 3.0

D. 3.25

E. 4.75

•

Alex deposited x dollars into a new account that earned 8 percent annual interest, compounded annually. One year later Alex deposited an additional x dollars into the account. If there were no other transactions and if the account contained w dollars at the end of two years, which of the following expresses x in terms of w?

A. $\frac{W}{1+1.08}$

B. $\frac{W}{1.08+1.16}$

C. $\frac{W}{1.16+1.24}$

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$1.16 + 1.24$

D. $\dfrac{W}{1.08 + (1.08)^2}$

E. $\dfrac{W}{(1.08)^2(1.08)^3}$

•

M is the sum of the reciprocals of the consecutive integers from 201 to 300, inclusive. Which of the following is true?

A. $\frac{1}{3} < M < \frac{1}{2}$

B. $\frac{1}{5} < M < \frac{1}{3}$

C. $\frac{1}{7} < M < \frac{1}{5}$

D. $\frac{1}{9} < M < \frac{1}{7}$

E. $\frac{1}{12} < M < \frac{1}{9}$

•

Working simultaneously at their respective constant rates, Machines A and B produce 800 nails in x hours. Working alone at its constant rate, Machine A produces 800 nails in y hours. In terms of x and y, how many hours does it take Machine B, working alone at its constant rate, to produce 800 nails?

A. $\dfrac{x}{x+y}$

B. $\dfrac{y}{x+y}$

C. $\dfrac{xy}{x+y}$

D. $\dfrac{xy}{x-y}$

E. $\dfrac{xy}{y-x}$

•

Carol purchased one basket of fruit consisting of 4 apples and 2 oranges and another basket of fruit consisting of 3 apples and 5 oranges. Carol is to select one piece of fruit at random from each of the two baskets. What is the probability that one of the two pieces of fruit selected will be an apple and the other will be an orange?

A. $\frac{1}{4}$

B. $\frac{1}{2}$

C. $\frac{1}{24}$

D. $\frac{5}{24}$

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E. $\frac{13}{24}$

•

Last year Brand X shoes were sold by dealers in 403 different regions worldwide, with an average (arithmetic mean) of 98 dealers per region. If last year these dealers sold an average of 2,488 pairs of Brand X shoes per dealer, which of the following is closest to the total number of pairs of Brand X shoes sold last year by the dealers worldwide?

A. 10^4

B. 10^5

C. 10^6

D. 10^7

E. 10^8

•

A positive integer is divisible by 3 if and only if the sum of its digits is divisible by 3. If the six-digit integer n is divisible by 3 and n is of the form $1k2,k24$, where k represents a digit that occurs twice, how many values could n have?

A. Two

B. Three

C. Four

D. Five

E. Ten

	Number of Marbles in Each of Three Bags	Percent of Marbles in Each Bag That Are Blue (to the nearest tenth)
Bag P	37	10.8%
Bag Q	x	66.7%
Bag R	32	50.0%

•

If $\frac{1}{3}$ of the total number of marbles in the three bags listed in the table above are blue, how many marbles are there in Bag Q?

A. 5

B. 9

C. 12

D. 23

E. 46

Age Category (in years)	Number of Employees
Less than 20	29
20–29	58
30–39	36
40–49	21
50–59	10
60–69	5

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70 and over	2

•

The table above gives the age categories of the 161 employees at Company X and the number of employees in each category. According to the table, if m is the median age, in years, of the employees at Company X, then m must satisfy which of the following?

A. $20 \leq m \leq 29$

B. $25 \leq m \leq 34$

C. $30 \leq m \leq 39$

D. $35 \leq m \leq 44$

E. $40 \leq m \leq 49$

•

If k and n are positive integers such that $n > k$, then $k! + (n-k) \cdot (k-1)!$ is equivalent to which of the following?

A. $k \cdot n!$

B. $k! \cdot n$

C. $(n-k)!$

D. $n \cdot (k+1)!$

E. $n \cdot (k-1)!$

•

Ron is 4 inches taller than Amy, and Barbara is 1 inch taller than Ron. If Barbara's height is 65 inches, what is the median height, in inches, of these three people?

A. 60

B. 61

C. 62

D. 63

E. 64

•

If x and y are positive numbers such that $x + y = 1$, which of the following could be the value of $100x + 200y$?

I. 80

II. 140

III. 199

A. II only

B. III only

C. I and II

D. I and III

E. II and III

•

If X is the hundredths digit in the decimal $0.1X$ and if Y is the thousandths digit in the decimal $0.02Y$, where X and Y are nonzero digits, which of the following is closest to the greatest possible value of $\frac{0.1X}{0.02Y}$?

A. 4

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B. 5

C. 6

D. 9

E. 10

•

If each of the 12 teams participating in a certain tournament plays exactly one game with each of the other teams, how many games will be played?

A. 144

B. 132

C. 66

D. 33

E. 23

•

For which of the following values of x is $\sqrt{1-\sqrt{2-\sqrt{x}}}$ not defined as a real number?

A. 1

B. 2

C. 3

D. 4

E. 5

•

when What is the remainder 3^{19} is divided by 10?

A. 1

B. 3

C. 5

D. 7

E. 9

•

Of 200 people surveyed, 80 percent own a cellular phone and 45 percent own a pager. If all 200 people surveyed own a cellular phone, or a pager, or both, what percent of those surveyed either do not own a cellular phone or do not own a pager?

A. 35%

B. 45%

C. 55%

D. 65%

E. 75%

•

At the end of the first quarter, the share price of a certain mutual fund was 20 percent higher than it was at the beginning of the year. At the end of the second quarter, the share price was 50 percent higher than it was at the beginning of the year. What was the percent increase in the share price from the end of the first quarter to the end of the second quarter?

A. 20%

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B. 25%

C. 30%

D. 33%

E. 40%

$$\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \ldots$$

•

In the sequence above, each term after the first is one-half the previous term. If x is the tenth term of the sequence, then x satisfies which of the following inequalities?

A. $0.1 < x < 1$

B. $0.01 < x < 0.1$

C. $0.001 < x < 0.01$

D. $0.0001 < x < 0.001$

E. $0.00001 < x < 0.0001$

•

If $xy + z = x(y + z)$, which of the following must be true?

A. $x = 0$ and $z = 0$

B. $x = 1$ and $y = 1$

C. $y = 1$ and $z = 0$

D. $x = 1$ or $y = 0$

E. $x = 1$ or $z = 0$

•

Two water pumps, working simultaneously at their respective constant rates, took exactly 4 hours to fill a certain swimming pool. If the constant rate of one pump was 1.5 times the constant rate of the other, how many hours would it have taken the faster pump to fill the pool if it had worked alone at its constant rate?

A. 5

B. $\frac{16}{3}$

C. $\frac{11}{2}$

D. 6

E. $\frac{20}{3}$

•

If $0 < x < 1$, what is the median of the values x, x^{-1}, x^2, $\sqrt{x}$, and x^3?

A. x

B. x^{-1}

C. x^2

D. $\sqrt{x}$

E. x^3

•

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The number of stamps that Kaye and Alberto had were in the ratio 5:3, respectively. After Kaye gave Alberto 10 of her stamps, the ratio of the number Kaye had to the number Alberto had was 7:5. As a result of this gift, Kaye had how many more stamps than Alberto?

A. 20

B. 30

C. 40

D. 60

E. 90

7.51	8.22	7.86	8.36
8.09	7.83	8.30	8.01
7.73	8.25	7.96	8.53

•

A vending machine is designed to dispense 8 ounces of coffee into a cup. After a test that recorded the number of ounces of coffee in each of 1,000 cups dispensed by the vending machine, the 12 listed amounts, in ounces, were selected from the data. If the 1,000 recorded amounts have a mean of 8.1 ounces and a standard deviation of 0.3 ounce, how many of the 12 listed amounts are within 1.5 standard deviations of the mean?

A. Four

B. Six

C. Nine

D. Ten

E. Eleven

•

In a certain quiz that consists of 10 questions, each question after the first is worth 4 points more than the preceding question. If the 10 questions on the quiz are worth a total of 360 points, how many points is the third question worth?

A. 18

B. 24

C. 26

D. 32

E. 44

•

A car traveling at a certain constant speed takes 2 seconds longer to travel 1 kilometer than it would take to travel 1 kilometer at 75 kilometers per hour. At what speed, in kilometers per hour, is the car traveling?

A. 71.5

B. 72

C. 72.5

D. 73

E. 73.5

•

If the speed of x meters per second is equivalent to the speed of y kilometers per hour, what is y in terms of x? (1 kilometer = 1,000 meters)

A. $\frac{5x}{18}$

B. $\frac{6x}{}$.

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B. $\frac{\ }{5}$.

C. $\frac{18x}{5}$

D. $60x$

E. $3,600,000x$

•

The function f is defined by $f\left(x\right) = -\frac{1}{x}$ for all nonzero numbers x. If $f\left(a\right) = -\frac{1}{2}$ and $f\left(ab\right) = \frac{1}{6}$, then $b =$

A. 3

B. $\frac{1}{3}$

C. $-\frac{1}{3}$

D. -3

E. -12

•

If $5^x - 5^{x-3} = (124)(5^y)$, what is y in terms of x?

A. x

B. $x - 6$

C. $x - 3$

D. $2x + 3$

E. $2x + 6$

•

The total of Company C's assets in 1994 was 300 percent greater than the total in 1993, which in turn was 400 percent greater than the total in 1992. If the total of Company C's assets in 1992 was N dollars, which of the following represents the total of Company C's assets, in dollars, in 1994?

A. $7N$

B. $8N$

C. $9N$

D. $12N$

E. $20N$

•

If $\frac{x}{|y|} = -1$, which of the following must be true?

A. $x = -y$

B. $x = y$

C. $x = y^2$

D. $x^2 = y^2$

E. $x^3 = y^3$

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E. $x^3 = y^3$

A	1	$2\sqrt{3}$
B	$\sqrt{6}$	C
$\sqrt{3}$	6	D

•

The table above is a 3 by 3 grid of 9 numbers, 5 of which are given. The 4 numbers that are not given are denoted by A, B, C, and D. The product of the numbers in each row and in each column is the same for all rows and columns. What is the value of the product $ABCD$?

A. 36

B. 72

C. $36\sqrt{2}$

D. $36\sqrt{3}$

E. $72\sqrt{6}$

•

In May, Mrs. Lee's earnings were 60 percent of the Lee family's total income. In June, Mrs. Lee earned 20 percent more than in May. If the rest of the family's income was the same both months, then in June, Mrs. Lee's earnings were approximately what percent of the Lee family's total income?

A. 64%

B. 68%

C. 72%

D. 76%

E. 80%

•

John would have reduced the time it took him to drive from his home to a certain store by $\frac{1}{3}$ if he had increased his average speed by 15 miles per hour. What was John's actual average speed, in miles per hour, when he drove from his home to the store?

A. 25

B. 30

C. 40

D. 45

E. 50

•

Which of the following inequalities has a solution set that, when graphed on a number line, is a single line segment of finite length?

A. $x^4 \geq 1$

B. $x^3 \leq 27$

C. $x^2 \geq 16$

D. $2 \leq |x| \leq 5$

E. $2 \leq 3x + 4 \leq 6$

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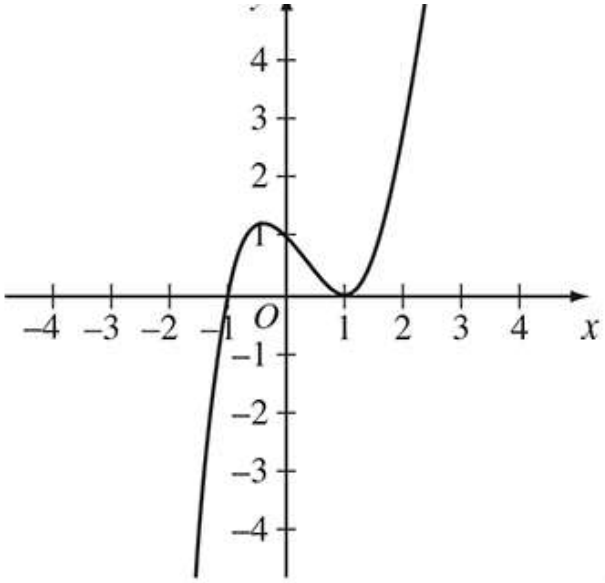

•

The figure shows the graph of $y = \left(x+1\right)(x-1)^2$ in the xy-plane. At how many points does the graph of $y = \left(x+1\right)(x-1)^2 + 2$ intercept the x-axis?

A. None

B. One

C. Two

D. Three

E. Four

Questions 177 to 272 — Difficulty: Hard

177. The cost to rent a small bus for a trip is x dollars, which is to be shared equally among the people taking the trip. If 10 people take the trip rather than 16, how many more dollars, in terms of x, will it cost per person?

A. $\frac{x}{6}$

B. $\frac{x}{10}$

C. $\frac{x}{16}$

D. $\frac{3x}{40}$

E. $\frac{3x}{80}$

178. Clarissa will create her summer reading list by randomly choosing 4 books from the 10 books approved for summer reading. She will list the books in the order in which they are chosen. How many different lists are possible?

A. 6

B. 40

C. 210

D. 5,040

E. 151,200

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179. If n is a positive integer and the product of all the integers from 1 to n, inclusive, is divisible by 990, what is the least possible value of n?

A. 8

B. 9

C. 10

D. 11

E. 12

180.

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